

Rectangles & Squares — Answers

Final answers with a one-line reason. For full methods, see the worked examples in the notes.

Objective

- Q1** **(b) Square.** Four right angles + four equal sides.
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- Q2** **(a) equal and bisect each other.** (They are perpendicular only in a square.)
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- Q3** **2.** Diagonals are equal: $2x + 1 = 3x - 1 \Rightarrow x = 2$.
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- Q4** **(c) Square.** Only the square has diagonals that are equal AND perpendicular.
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- Q5** **True.** A square has four right angles, so it is a rectangle.
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- Q6** **(a) Both true, R explains A.** A rectangle has equal diagonals, and a square is a rectangle.
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Short answer

- Q7** $7\sqrt{2} \approx 9.9$ cm. Diagonal of a square = side $\times \sqrt{2}$.
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- Q8** **15 cm.** $\sqrt{(9^2 + 12^2)} = \sqrt{225} = 15$.
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- Q9** **OC = OB = OD = 5 cm; AC = 10 cm.** Diagonals are equal and bisect each other, so all half-diagonals are equal.
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Long answer

- Q10** **$x = 4$; each diagonal = 17 cm.** $5x - 3 = 3x + 5 \Rightarrow 2x = 8 \Rightarrow x = 4$; $5(4) - 3 = 17$.
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- Q11** **Draw the two 6 cm diagonals so they bisect each other at 90° , then join the four endpoints.** A square's diagonals are equal and meet at 90° .
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- Q12** **(i) and (iv) are always true.** Rectangle diagonals are equal and bisect each other; they are not perpendicular and the sides are not all equal (unless it is a square).
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Total: 21 marks.